Pricing across two markets with an AI-assisted workflow

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1. Summary

	Result
Week 1 (developed market)	Private leaderboard rank 3 of 54 teams, pooled Tweedie D2 0.13864 (top team 0.13923); selected file `blend_v10_perline.csv`. Public rank 14: the public order inside the top group did not predict the private order (section 6).
Week 2 (emerging market), public leaderboard	0.16458 for the final ensemble (selected on Kaggle), from 0.16077 for the first (twelve submissions; the last eight, 0.16450 to 0.16474, differ by less than the noise of the public board; the best single file, 0.16474, is within 0.00016 of the final). **Week 2 private leaderboard: rank 2; Week 1: rank 3 of 54.**
Week 2 local validation (two emerging years, both time directions)	Pooled D2 0.1405 forward (calibrate 2023, score 2024) and 0.1268 reverse for the final ensemble (0.1397 / 0.1271 before ensembling), against 0.1110 forward for a model fitted on the emerging data alone.

Approach in one paragraph. The emerging book has two years and about 1,000 claims per property line, too little to learn its own relationships. We therefore keep the developed market's frequency-by-severity model (trained on 2021-24 with 2.4 million policy rows) for the *relationships between rating factors and cost*, restricted to variables both markets record, and re-estimate only the *level* of each peril and one market-specific rating factor (gender) on the emerging data, with credibility weighting. Every change was judged on the two emerging years in both time directions and on paired bootstrap differences; more than twenty plausible ideas were rejected on that evidence (section 4).

2. What changed in the new market

Topic	Developed	Emerging	Consequence
Motor claim frequency / mean cost	reference	+34% / -42% (median cost -37%)	one level factor is not enough: perils move differently
Perils (frequency ratio emerging /		glass-hail 1.5-1.8, third-party property	stable between the two emerging years

Topic	Developed	Emerging	Consequence
developed)		1.4, bodily injury 1.4-1.5, escape of water 0.65-0.7, storm 1.3-1.7	for the large perils
Severity scale (median ratio)		motor ~0.6, buildings ~0.7, contents ~0.75, fire ~1.0	a severity factor per peril, credibility-weighted
Book	15% new business in 2025	**76% new business** in the 2025 test, 14% in training	the new-business relativity must come from the developed market
Recorded information	exact age, mileage, telematics, credit band	age bands, gender, free-text occupation, local construction codes; no history before 2023	shared variables only; harmonisation layer
Claim reporting		same lag pattern and open share (3.3% / 19.7% open for 2023 / 2024 vs 3.2% / 20.2%)	the developed completeness table is reused

Data quality (all in `docs/part2/DATA_REVIEW.md`): the Part 1 duplicates and placeholder dates are absent, but the same families appear in new forms: `sum_insured` multiplied by 1,000 on 0.2% of rows including 239 test rows, occupation spelled 81 ways, local construction codes, 36 claims dated before their policy, a corrupted exposure column (0.5% of rows), 12% of selected fields missing completely at random, and the customer-service-contacts column that is generated by the outcome (correlation 0.77 with claim count in training, half the level in the test) and is never used. An independent review by a second agent that saw only the raw tables found the same defects and two more (10% of developed motor customers missing from the customer table; emerging vehicle ages start at 3), both verified.

3. Method

1. **Developed model on shared variables** (`src/iaa/market.py`): exact ages are cut into the emerging bands, occupation is normalised to 20 values plus a self-employed flag, region, mileage, telematics and credit band are left out. LightGBM frequency (Poisson, exposure offset) by severity (Gamma on claims capped at the 99.9% quantile, trended to the predicted year, excess load added back) per peril, completeness offset for immature years, repaired data (`src/iaa/cleaning.py`, fail-fast count bands).
2. **One-year claim history** (`src/iaa/p2_features.py`): the policy's own claims of the previous year, the claims at the address and the customer's claims over all lines. Windows are exactly one year because the emerging book starts in 2023; the first year of each market is *unknown*, not zero. Renewals with a claim last year were 27% above the

model (actual / expected 1.195 vs 0.944 in motor); the gain is +0.0023 forward and +0.0021 reverse.

3. **Base blend** (the Part 1 recipe): 50% frequency-by-severity GBM, 25% direct pure-premium Tweedie GBM (two settings), 25% two neural nets (TabM and TabM-PLE), all with the history inputs.
4. **Level on the emerging book** (`src/iaa/transfer.py`): per peril a frequency factor from claim counts and a severity factor from claim means winsorised at the 99% quantile of each market, both shrunk towards the line value by credibility (k = 20 and 30 claims; buildings frequency k = 400 because its peril ratios swing between the two years: fire 2.3x then 0.84x); fitted on both emerging years; **no final scaling to the training total**, which copies the large-loss noise of the two years into the level.
5. **Gender** (allowed for pricing in the emerging market by the competition; regulatory status elsewhere must be checked): one credibility-weighted factor per line, motor male / female 1.118 / 0.880. It is stable in both emerging years (motor frequency +29% and +34%); the file without it is one switch away.

4. Validation, and what was rejected

Design. Fold E is the only design the data allows: calibrate on one emerging year, score the other, in both directions ("forward" 2023 -> 2024, "reverse" 2024 -> 2023), because with two years a single split mistakes noise for signal (buildings moves by a factor of 1.5 between the years). A change is adopted only if neither direction is worse and the combined gain is clearly positive; paired customer-cluster bootstrap on the same rows.

Step	Forward	Reverse	Public
Emerging-only GBM	0.1110		
Developed model, level closed to the training total, gender	0.1256	0.1162	0.16077
+ robust level (no closure)	0.1281	0.1176	0.16314
+ nets (TabM, TabM-PLE) 50%	0.1289	0.1220	0.16336
+ one-year claim history	0.1313	0.1229	
+ direct Tweedie part (25%)	0.1336	0.1237	0.16450
+ buildings frequency credibility 400	0.1397	0.1271	0.16474
+ ensemble of 5 tree bases, 7 nets and 4 Tweedie fits (final, file p2_t11)	**0.1405**	**0.1268**	**0.16458**

The ensemble averages members with equal weights inside each group (single-seed and 3-seed fits, 4- and 8-leaf trees, three net families); it is variance reduction, not a tuned gain: the public differences among the last eight candidates are 0.00024 or less (a smaller ensemble of 3 tree bases, 6 nets and 4 Tweedie fits scored 0.1408 / 0.1272 locally and 0.16462 publicly; the larger one was selected for its wider composition). Checks that bound what more tuning could give: a simulation of the level method with a known truth (400 simulated books per scenario; the robust level beats a plain line closure in every scenario, RESULTS 76-78) and a ceiling analysis (a perfect line level would add at most 0.005 D2, RESULTS 84); a search for remaining feature structure found none beyond the gender factor (973 levels, RESULTS 79). Local figures are on the two emerging training years, public figures on the 2025 test rows (a different population: 76% new business); only the ordering is comparable.

Rejected with evidence (`docs/workflow/DECISIONS.md`, `docs/experiments/RESULTS.md`): emerging-only models, stacking and blends with them, a new-business multiplier (estimated 1.57 in 2023, 0.9 in 2024), region and other emerging-only features, age / occupation / excess / no-claims-bonus / cover-type corrections beyond gender, frequency-severity split calibration, domain-shift weights, a model without the insurer's own premium variables, the construction code MIX read as timber, brick or stone, engineered features (market-relative premium, percentile ranks, lines held, ratios), other LightGBM settings (leaves, learning rate, leaf size), three-seed averaging, a hail-slope correction (the developed model under-reacts to hail between addresses, but the correction fails in reverse), spread and exposure post-corrections. The pattern is consistent: gains that appear in one direction and not in the other sit in buildings, the line with ~1,000 claims, and are validation noise.

5. Uncertainty

Source	Effect	Treatment
Buildings level (two years: fire 49 vs 21 claims)	+-20% moves the validation score by +0.0069 / -0.0086 (forward) and -0.0112 / +0.0057 (reverse)	stated risk; the level uses both years, buildings peril ratios taken from the developed model
New-business relativity (80% of the predicted total)	observed / predicted new-business-to-renewal frequency 1.02-1.15 in motor, noisy elsewhere; +-20% moves the score by < 0.004	carried over from the developed market (motor +45% in both markets)
History does not exist for new business	on a validation set reweighted to 76% new business the gain of the full model over the one without history inputs shrinks to +0.0038 forward and -0.0009 reverse	gain expected on the 24% renewals only
Motor and contents level	+-20% moves the score by < 0.006	well pinned

6. Effective use of AI, and its verification

- **Where AI worked:** reading the competition pages, profiling both markets, writing the harmonisation, transfer and calibration code test-first (125 automated tests), running experiments, drafting documents. 11 helper sessions in separate containers ran the hours-long refits in parallel (nets, GBM variants) and an independent agent compared the two markets from the raw tables without seeing our results.
- **Verification, with examples of what it caught:** (i) three unsupported statements in the first draft of the data review (an exposure explanation, invented new-business counts, a leak pattern) were replaced by measured values before commit; (ii) the blind review's suggestion that the developed model over-spreads risk was tested and rejected (consistent in one direction only); (iii) a helper correctly refused a stop instruction that was not from the lead, which we then handled by archiving the session; (iv) every number in the documents traces to a command or a log line (`RESULTS.md` sections 45-65).
- **Two more catches in the last days:** (iv) our first simulation of the level method showed every estimator biased low on buildings; the plain line closure showed the same bias, which led to a defect in the simulator itself (claims drawn from all years, compared with one year), and the wrong conclusion was withdrawn before it reached any decision; (v) a net we built and submitted had already been rejected in Part 1 on the large developed folds, and the public score agreed with the old result; we now screen ranking ideas on the developed folds first and search the record before building (`docs/workflow/AI_USAGE.md`).
- **Part 1 shake-up as a lesson** (`docs/experiments/PART1_SHAKEUP.md`, figure `part1_public_vs_private.png`): public and private scores correlate 0.997 across 51 teams but 0.20 among the public top 25; the top ten by public ended at private ranks 1-27. We therefore select by local two-direction validation, not by public score differences below about 0.001.

7. Trust, validation, guardrails

Automated: 125 tests on every push (CI), fail-fast data bands for the repairs, a leak alarm (pooled D2 above 0.35 stops a run; `customer_service_contacts` excluded in both markets), a reproducibility assertion (the developed model re-run reproduces its first run to 1e-6), submission-format validation, scenario analysis (`scripts/p2_scenarios.py`).
Governance: 83 numbered decisions with evidence, a model card (`docs/part2/MODEL_CARD.md`) with monitoring triggers (actual / expected by line and by new-business flag; escalate outside 0.85-1.15), a one-command reproduction (`scripts/reproduce_part2.sh`), human sign-off by the project lead on every submission and decision (reviewer and approver names: to be filled in by the team). Standing rules: no forum posts, no leaderboard probing (D21), fixed validation design.

8. From risk premium to office premium (20 reference policies)

Office premium = (risk + 12% x risk for property + 85 x exposure) / (1 - 15% variable - commission 20% broker or 0% direct - 8% tax - 8% margin),
`src/iaa/office_premium.py`; full table

`docs/part2/office_premium_reference.csv`. Example rows (CU; risk premium from the latest candidate):

Policy	Line, channel	Exposure	Risk	Catre.	Fixed	Variable	Commission	Tax	Margin	**Office**	Written
M00161473	motor, broker	0.96	1,207	0	82	395	526	210	210	**2,630**	1,132
M00185311	motor, direct	0.96	930	0	82	220	0	117	117	**1,467**	872
M00129413	motor, broker	0.33	123	0	28	46	62	25	25	**309**	354

Median risk premium of the 20 policies is 0.88 of the written premium and the median office premium 2.5 times the written premium: the current premiums roughly equal expected claims before any expense, tax or margin, so in the emerging market they would need to rise sharply (broker business carries 51% of premium in loadings). The fixed expense is pro-rated by exposure and no reference policy is on the comparison-site channel, whose commission is blank in the table (an explicit assumption would be needed).

9. Limitations

Two emerging years; 76% of the test book is new business and has no claim history; the buildings level is uncertain by a factor that the validation cannot narrow; gender rating may not be permissible outside this competition; the construction code MIX and the developed claim-reporting pattern are assumptions that tested neutral; the public leaderboard score (0.16474) is on a different population from our validation and is used only as a level check.

Appendix (not graded): where to look

`docs/part2/DATA_REVIEW.md` (data), `VARIABLES.md` (every variable by market and how it is treated), `EXPLORATION.md` and `SECOND_OPINION.md` (differences and the blind review), `PART2_STRATEGY.md` (plan), `MODEL_CARD.md`, `EVIDENCE_MAP.md` (rubric to evidence), `docs/experiments/RESULTS.md` (all results), `docs/workflow/DECISIONS.md` and `AI_USAGE.md`.